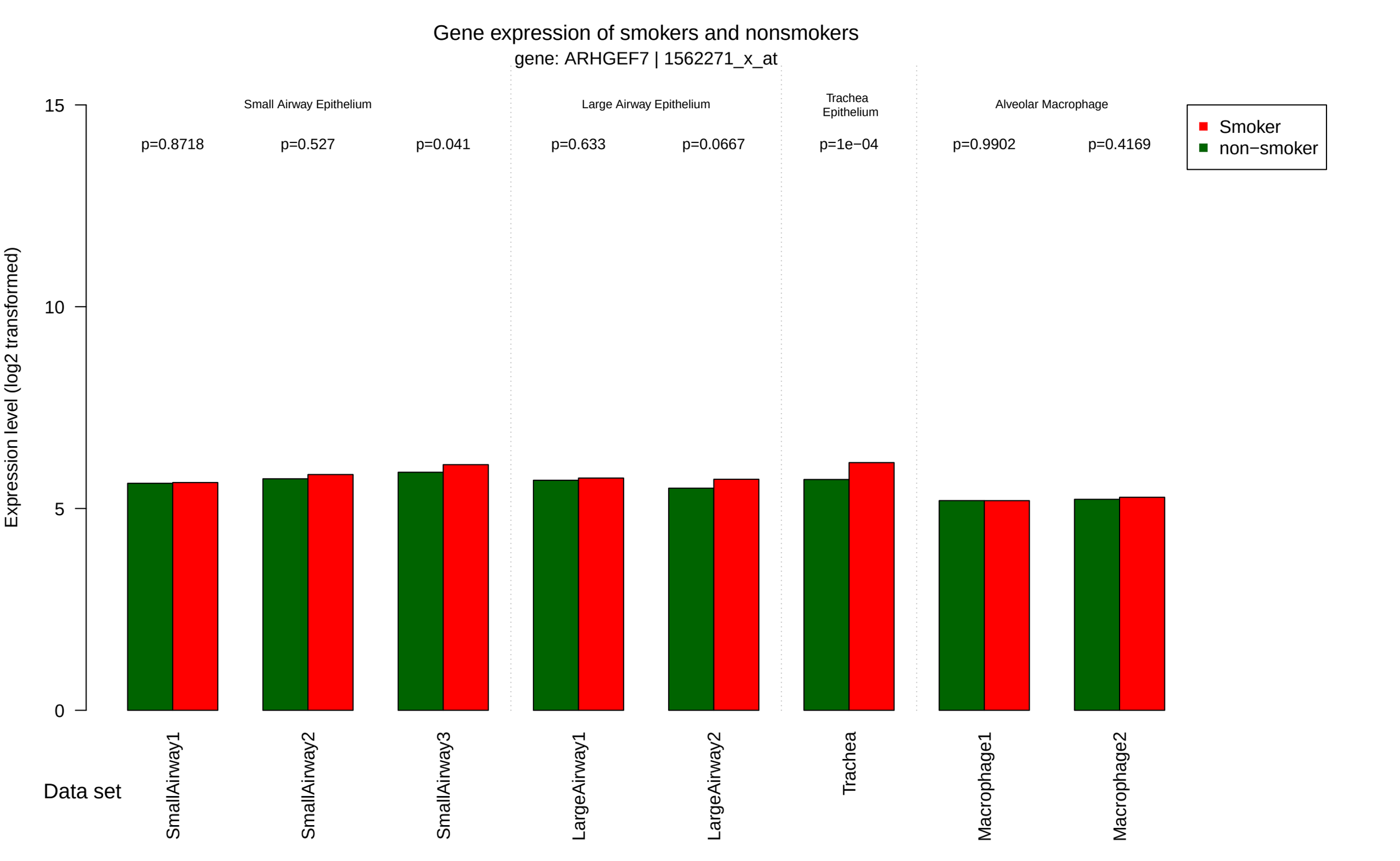
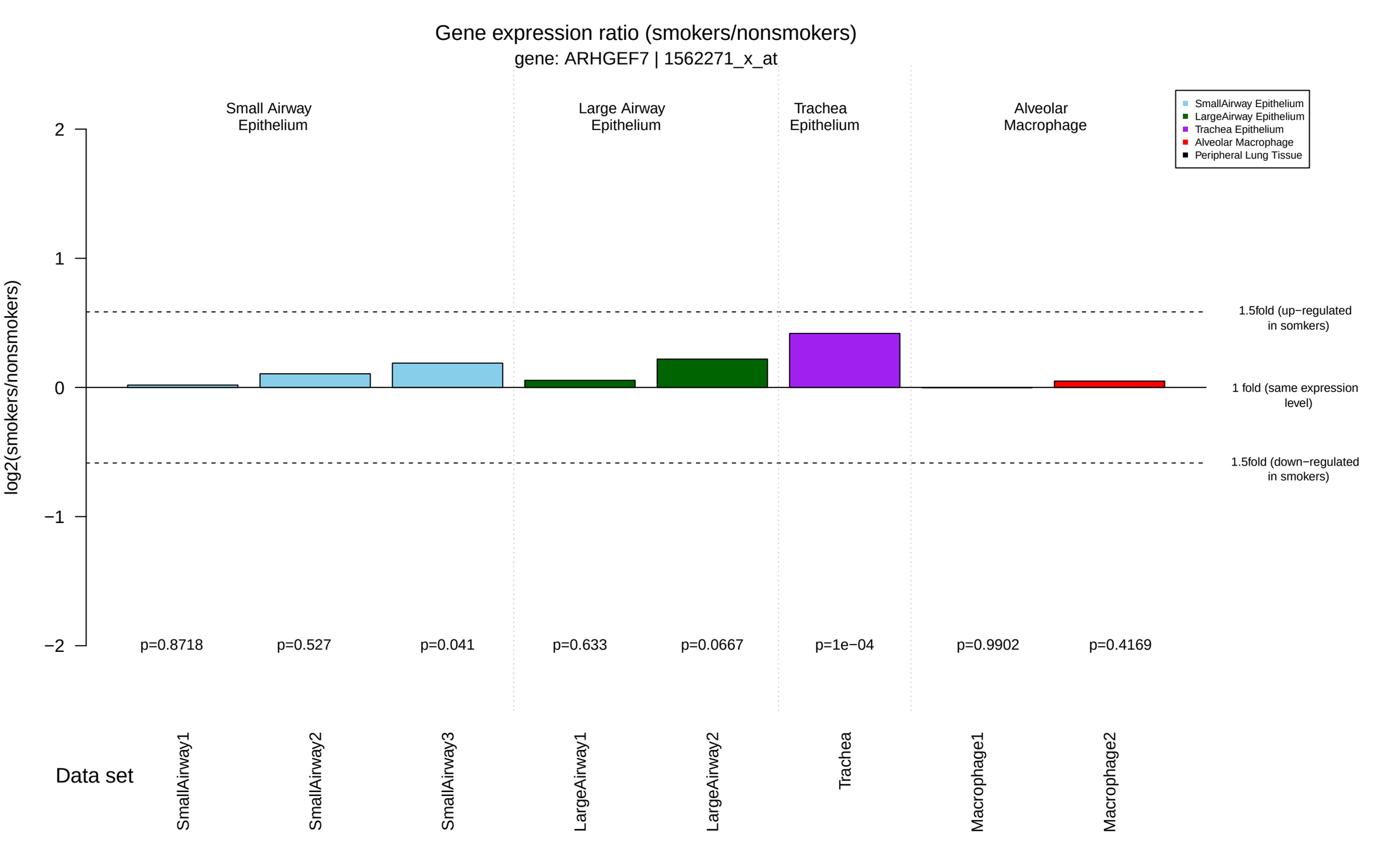
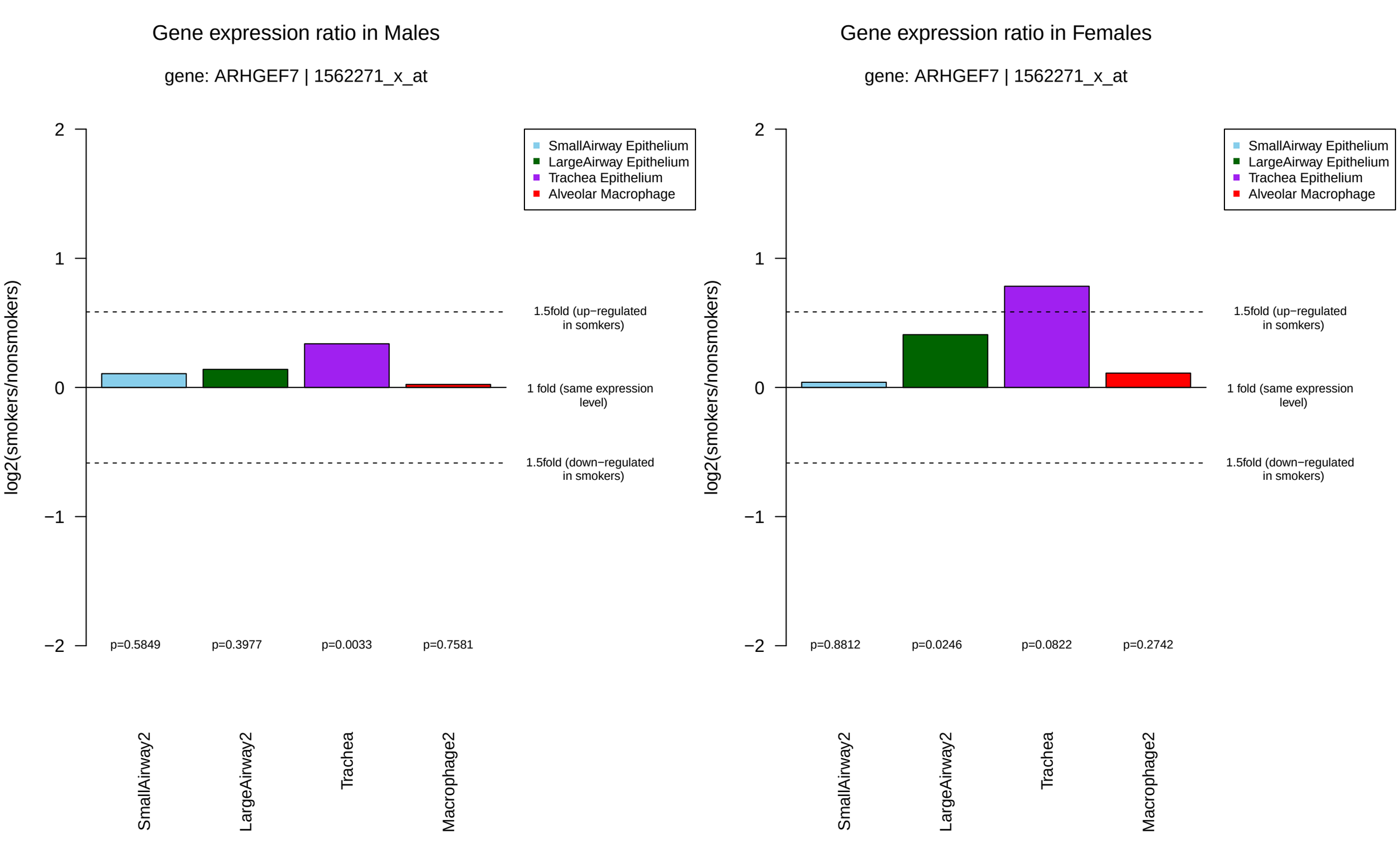
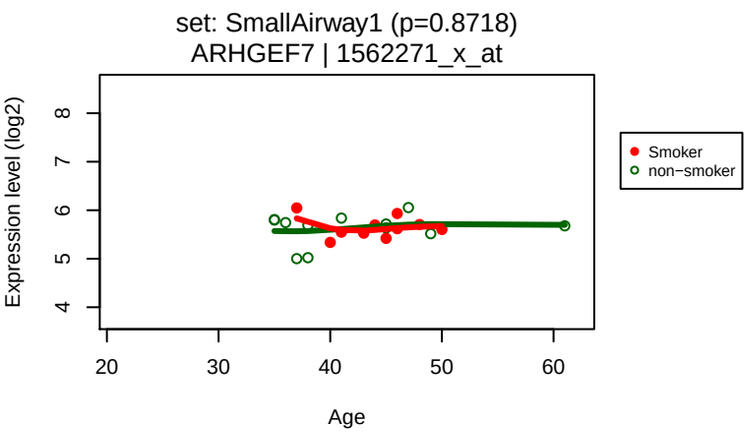
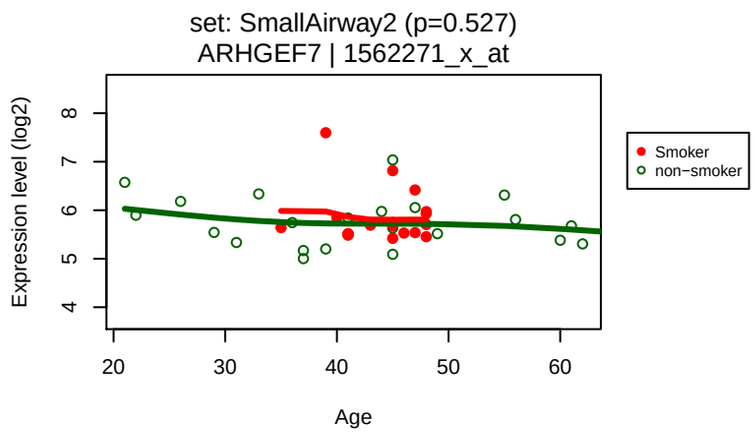
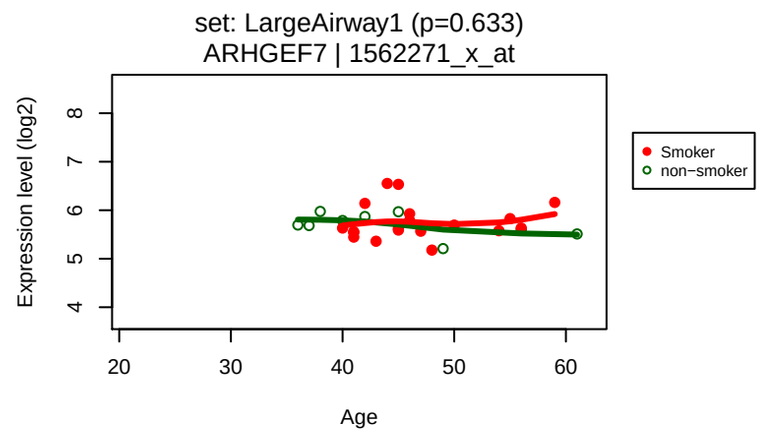

Gene expression of smokers and nonsmokers
gene: ARHGEF7 | 1562271_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8718
p=0.527
p=0.041
p=0.633
p=0.0667
p=1e−04
p=0.9902
p=0.4169
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

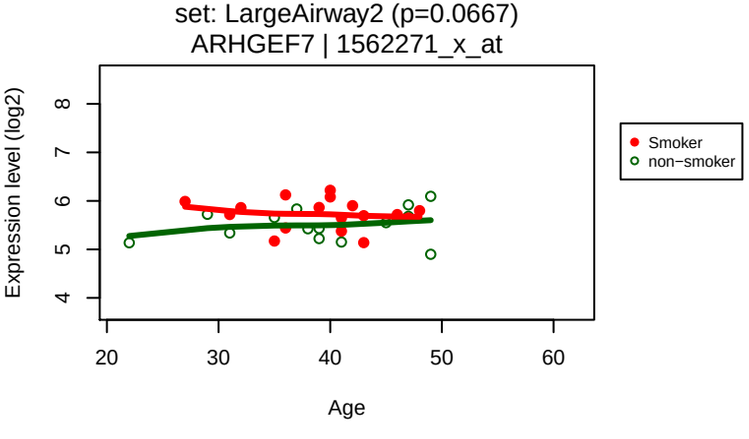

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF7 | 1562271_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8718
p=0.527
p=0.041
p=0.633
p=0.0667
p=1e−04
p=0.9902
p=0.4169
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

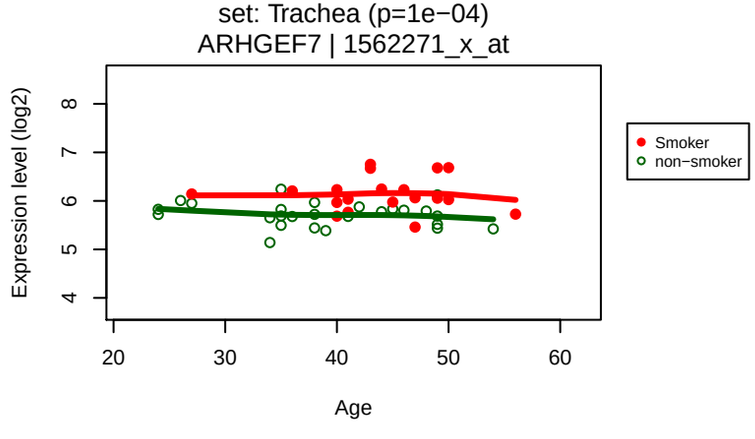

Gene expression ratio in Males
gene: ARHGEF7 | 1562271_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5849
p=0.3977
p=0.0033
p=0.7581
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF7 | 1562271_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8812
p=0.0246
p=0.0822
p=0.2742
SmallAirway2
LargeAirway2
Trachea
Macrophage2

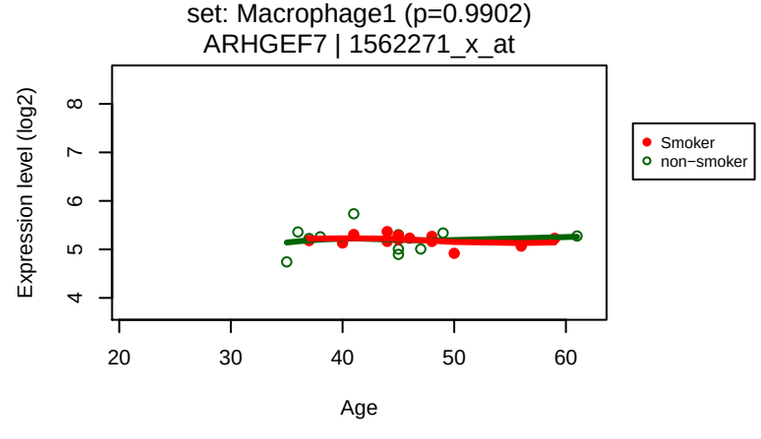

set: SmallAirway1 (p=0.8718)
ARHGEF7 | 1562271_x_at
Expression level (log2)
Age
Smoker
non−smoker

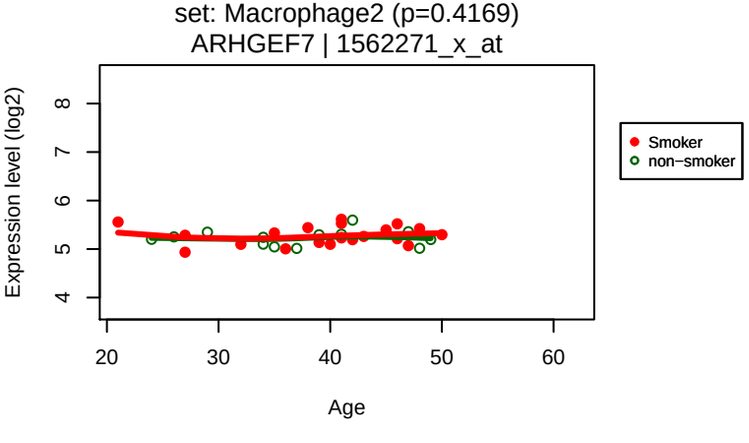

set: SmallAirway2 (p=0.527)
ARHGEF7 | 1562271_x_at
Expression level (log2)
Age
Smoker
non−smoker

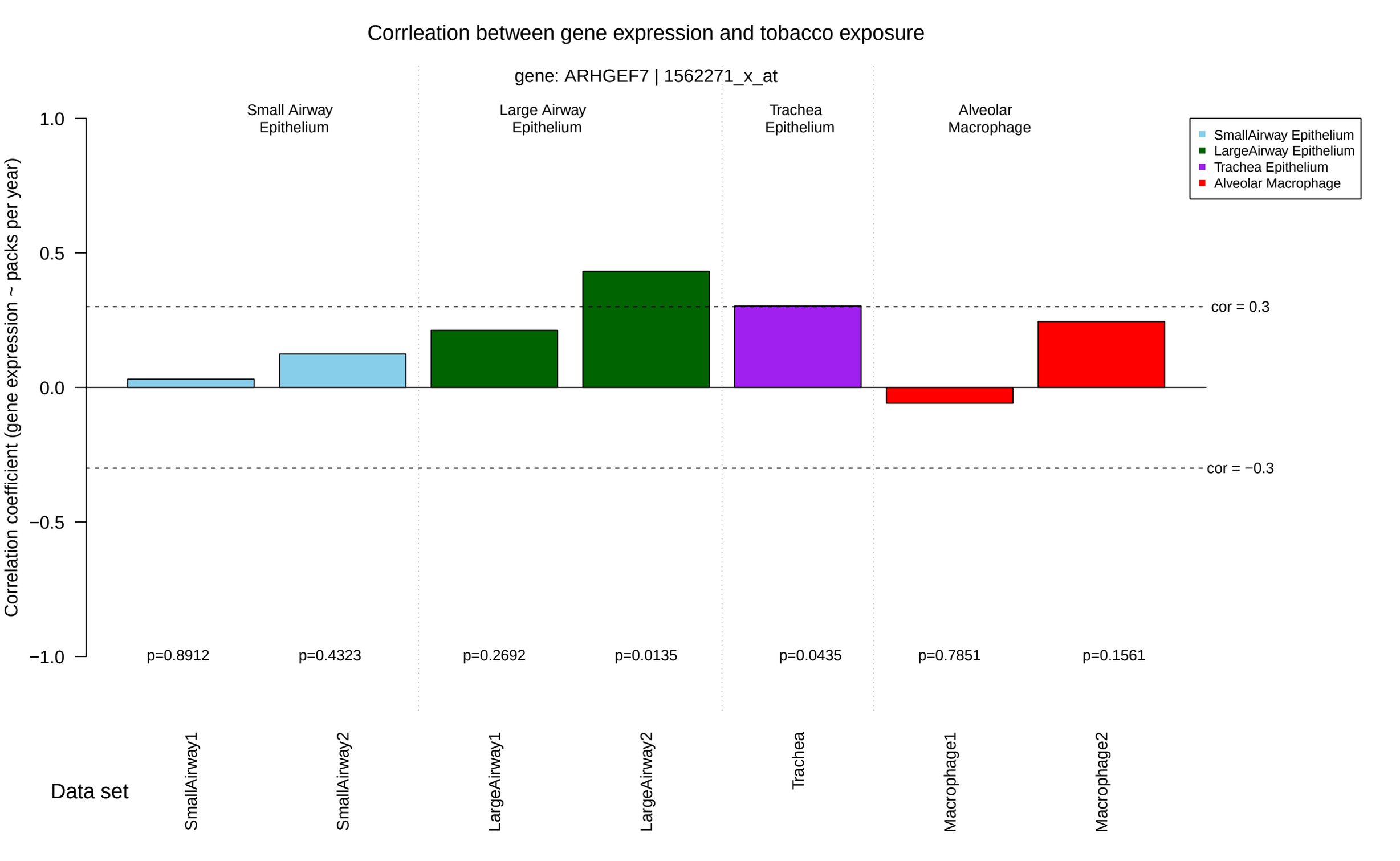

set: LargeAirway1 (p=0.633)
ARHGEF7 | 1562271_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0667)
ARHGEF7 | 1562271_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
ARHGEF7 | 1562271_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9902)
ARHGEF7 | 1562271_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4169)
ARHGEF7 | 1562271_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF7 | 1562271_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8912
p=0.4323
p=0.2692
p=0.0135
p=0.0435
p=0.7851
p=0.1561
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2